

# Amir Hossein Ansari

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 Amir Hossein Ansari |  Amir Hossein Ansari

Tehran, Iran

## EDUCATION

### • Sharif University of Technology (SUT)

Sep. 2022- Dec. 2024

Master of Science | Computer Engineering

Tehran, Iran

◦ CGPA: 18.80/20.00

#### My Master's Thesis

**Topic:** Develop a hybrid fault tolerance technique for improving the reliability of embedded systems while simultaneously reducing peak power consumption and satisfying timing constraints.

**Some key words:** Power consumption, Hybrid fault-tolerant technique, Scheduling, Multi-core embedded systems.

**Supervisor:** Dr. Alireza Ejlali

### • Shahed University

Sep. 2018 – Sep. 2022

Bachelor of Science | Electrical Engineering

Tehran, Iran

◦ CGPA: 17.51/20.00

#### My Bachelor's Thesis

**Topic:** Implementation of ARM Cortex-M1 Processor on FPGA with Embedded System Applications

**Some key words:** FPGA, ARM Cortex-M1, Verilog/VHDL, Xilinx Vivado, Embedded Systems, Digital Design

**Supervisor:** Dr. Mohsen Jalali

## PROJECTS

### • Dynamic Task Scheduling and Optimization in Heterogeneous Systems Using GNN and NSGA-III Jul-Aug 2025

Tools: Python, TensorFlow, PyTorch

◦ Designed a dynamic scheduling framework optimizing energy efficiency, execution time, and task migration.

### • Peak-Power-Aware Hybrid Fault-Tolerant Technique in Multicore Embedded Systems Sep. 2022- Dec. 2024

Tools: gem5, McPAT, Hotspot, MiBench and PARSEC Benchmark Suite, Python

◦ Proposed TAFT, Pilot, and HAMLET methods, reducing energy and peak power consumption while improving schedulability and reliability.

### • ASIC Implementation of Array-Based Complex Matrix Multiplier Spring 2023

Tools: Synopsys Design Compiler, ModelSim, SoC Encounter, VHDL, HSPICE

◦ Designed a low-power array multiplier with configurable precision and optimized PDP in a 45nm process.

### • ISCAS Combinational Circuit Simulator with Automated Test Vector Generation Spring 2023

Tools: Python, Verilog, PODEM Algorithm, Fault Simulation

◦ Developed a simulator for ISCAS circuits with automatic test vector generation and fault injection analysis.

### • FPGA-Based Implementation of ARM Cortex-M1 Processor Spring 2022 - Summer 2022

Tools: Xilinx Vivado, Verilog/VHDL, Xilinx SDK, ARM Keil

◦ Implemented an FPGA-based ARM Cortex-M1 processor with optimized pipeline and low-latency interrupt handling.

### • Analysis and Implementation of RF Amplifier Circuits with PSpice Simulation Fall 2021

Tools: PSpice, MATLAB

◦ Designed and simulated an RF amplifier circuit, optimizing gain, bandwidth, and harmonic distortion reduction.

## RESEARCH EXPERIENCE

### ◦ Embedded Systems Research Laboratory

Sep. 2022 – Present

Supervisor: Dr. Alireza Ajlali

**Research Focus:** Power-Aware and Fault-Tolerant Techniques in Multicore Embedded Systems

- \* Developed TAFT, a thermal-aware hybrid fault-tolerant technique, achieving 41.8% energy savings and 13.5% peak power reduction, while improving system schedulability.
- \* Designed Pilot, a power-aware fault-tolerance framework, reducing peak power by 34.2% on average and improving QoS by 58.1%, ensuring system reliability under fault conditions.
- \* Implemented HAMLET, a heterogeneous adaptive task mapping and scheduling approach, integrating genetic algorithms, LSTM-based temperature prediction, and DVFS/DPM techniques to enhance energy efficiency and reliability.
- \* Conducted extensive simulations using gem5, McPAT, and HotSpot, analyzing the impact of thermal constraints on fault-tolerant scheduling.

**Skills Acquired:** Fault-tolerant system design, real-time scheduling, power-aware computing, machine learning for embedded systems, multi-core architectures; programming languages and tools: C/C++, Python, gem5, McPAT, HotSpot.

## RESEARCH INTERESTS

- Embedded Machine Learning
- Fault-Tolerant System Design
- Low-Power System Design
- Cyber-Physical Systems
- Real-Time Systems
- Internet of Things (IoT)

## PUBLICATION

- [1] **A. H. Ansari**, M. Esnaashari, M. Ansari, S. Safari, A. Ejlali, and J. Henkel, "Pilot: Power-Aware Hybrid Fault Tolerance in Multi-core Embedded Systems," *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 2026. 
- [2] **A. H. Ansari**, M. Ansari, A. Ejlali, and J. Henkel, "HAMLET: Heterogeneous Adaptive Mapping and Low-Energy Task Scheduling for Multi-core Embedded Systems," *IEEE Internet of Things Journal (IoT)*, 2026. 
- [3] **A. H. Ansari**, M. Ansari, S. Safari, A. Ejlali, and J. Henkel, "GLEAM: A Graph-Learning Enhanced Adaptive Metaheuristic for Power-Aware Scheduling on Heterogeneous Cyber-Physical Systems," in *2026 Design, Automation & Test in Europe Conference (DATE)*, 2026. 
- [4] **A. H. Ansari**, M. Ansari, and A. Ejlali, "TAFT: Thermal-Aware Hybrid Fault Tolerant Technique for Multicore Embedded Systems," *IEEE Embedded Systems Letters*, 2024. 
- [5] A. M. Rasouli, **A. H. Ansari**, M. Esnaashari, S. Safari, and M. Ansari, "A Two-State Energy-Efficient Reliability-Aware Strategy in Fault-Tolerant Embedded Systems," *IEEE Embedded Systems Letters*, 2025. 

## SUBMISSIONS

- [1] M. Hajilou, **A. H. Ansari**, A. Najar, M. H. Keshavarzi, M. Ansari, and J. Henkel, "Energy-Aware Graph Neural and Complex Network Approach in Fault-Tolerant Cyber-Physical Systems," Submitted to *IEEE Transactions on Parallel and Distributed Systems (TPDS)*.
- [2] M. Taghavi, **A. H. Ansari**, M. Esnaashari, M. Ansari, and A. Ejlali, "ETAGS: Energy and Temperature-Aware Graph-Based Scheduling in Heterogeneous Real-Time Embedded Systems," Submitted to *IEEE Transactions on Emerging Topics in Computing (TETC)*.
- [3] M. Esnaashari, K. Kianian, **A. H. Ansari**, M. Ansari, and S. Safari, "MERITS: Multiobjective Optimization of Energy-Efficient Real-Time Jobs Scheduling for Fault-Tolerant Configurable Industrial Systems," Submitted to *IEEE Transactions on Industrial Informatics (TII)*.

## TEACHING EXPERIENCE

- **Teaching Assistant** Sharif University of Technology
  - \* **Graduate Courses**
    - Low Power System Design, Dr. Alireza Ejlali Spring 2024, Spring 2026
    - Fault-Tolerant System Design, Dr. Alireza Ejlali Fall 2023, Fall 2024
    - Cyber-Physical Systems, Dr. Mohsen Ansari Fall 2023, Fall 2024, Fall 2025
    - System-on-Chip Design, Dr. Mohsen Ansari Spring 2024
  - \* **Undergraduate Courses**
    - Digital Systems Design, Dr. Alireza Ejlali Spring 2023, Spring 2024
    - Embedded Systems, Dr. Mohsen Ansari Years 2022 – 2026
    - Real-Time Systems, Dr. Sepideh Safari Years 2023 – 2026
    - Digital Systems Design, Dr. Amin Foshati Spring 2024
- **Lab Instructor** Sharif University of Technology
  - \* Digital Systems Design Lab, Dr. Alireza Ejlali Spring 2024, Summer 2024
  - \* Logic Circuits Lab, Dr. Mohsen Ansari Spring, Summer, and Fall 2023-2024
  - \* Logic Circuits Lab, Dr. Shaahin Hessabi Summer 2024

## SKILLS

- **Programming Languages:** C, C++, Python, MATLAB, Latex
- **Hardware Description Languages:** Verilog, VHDL
- **Hardware & Electronics:** STM32 Microcontrollers, AVR Microcontrollers, Arduino, PCB Design, Raspberry Pi, Proteus Design Suite, PSpice, Keil  $\mu$ Vision
- **Design & Simulation Tools:** Xilinx ISE, Gem5, McPat, Hotspot, ModelSim, Synopsys Design Compiler, SOC Encounter
- **Operating Systems:** Windows, Linux

## PROFESSIONAL SERVICES

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### • Reviewer – Journals

- IEEE Internet of Things Journal (IoT)
- IEEE Open Journal of The Computer Society

### • Sub-Reviewer – Journals

- IEEE Transactions on Parallel and Distributed Systems (TPDS)
- IEEE Transactions on Computers (TC)
- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
- IEEE Transactions on Sustainable Computing (TSUSC)
- IEEE Embedded Systems Letters
- ACM Transactions on Design Automation of Electronic Systems (TODAES)
- ACM Transactions on Embedded Computing Systems (TECS)
- ACM Transactions on Sensor Networks (TSN)

### • Sub-Reviewer – Conferences

- Euromicro Digital System Design Conference (DSD 2024)
- Asia and South Pacific Design Automation Conference (ASP-DAC 2024, ASP-DAC 2025)
- 5th CPSSI International Symposium on Cyber-Physical Systems (Application and Theory) (CPSAT 2024)
- IEEE 14th International Symposium on Industrial Embedded Systems (SIES 2024)

## HONORS AND AWARDS

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- **Ranked 3rd** among all B.E. students in the Electrical Engineering Department at Shahed University.
- **Ranked 21st** among approximately 15,000 students in the Master's National Entrance Exam in Iran.

## RELATED COURSES

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- **Master of Science (out of 20):** Low Power System Design (20), Cyber-Physical Systems(19.6), Seminar (19.9), Advanced VLSI Design (19.5), System-on-Chip Design (19.2), Fault-Tolerant System Design (19)
- **Bachelor of Science (out of 20):** Communication Systems (20), Electrical Engineering Co-Op (20), Electrical Engineering Project (19.75), Engineering Probability and Statistics (19.6), Digital Communication (19), Microprocessor (19)

## ADDITIONAL INFORMATION

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**Languages:** Persian (Native), English

## REFERENCES

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1. **Dr. Alireza Ejlali**   
Department: Department of Computer Engineering, Sharif University of Technology, Tehran, Iran  
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*Relationship: My Master's Supervisor*
2. **Dr. Shaahin Hessabi**   
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*Relationship: My instructor for Advanced VLSI and Testability Courses*
3. **Dr. Mohsen Ansari**   
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*Relationship: My Papers' Collaborator Supervisor*
4. **Dr. Sepideh Safari**   
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*Relationship: My Pilot Paper' Collaborator Supervisor*